



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 6 · STANDARD 6.AT.7

Properties of Operations

Lesson 6-4 · Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can use the commutative, associative, and identity properties to rewrite expressions.

Language Objective: I can explain my reasoning using the words commutative property, associative property, and identity property.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Commutative Property

Associative Property

Identity Property

Property

Distributive Property



Tap any word to see what it means and a picture.

1

The rule that order does not change a sum or product, like $a + b = b + a$, is the

2

The rule that grouping does not change a sum or product, like $(a + b) + c = a + (b + c)$, is the

3

The rule that adding 0 or multiplying by 1 keeps a number the same is the

4

A rule that is always true in math is a

5

The rule that lets you multiply a number across a sum, like $a(b + c) = a \times b + a \times c$, is the

Reflect — Exit Ticket

Which property is shown? $(8 + 5) + 2 = 8 + (5 + 2)$

- A. Associative Property
- B. Commutative Property
- C. Identity Property
- D. Distributive Property

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Commutative Property
2. **Notes 2:** Associative Property
3. **Notes 3:** Identity Property
4. **Notes 4:** Property
5. **Notes 5:** Distributive Property
6. **Watch for this mistake:** *A common mistake in Properties of Operations is skipping the key idea: "Changing the order or the grouping of numbers being added or multiplied does not change the answer." — always check your work against this rule before you submit.*
7. **Try It 1:** A. Associative Property — *The grouping (parentheses) changed but the order of the numbers stayed the same, so this is the Associative Property.*
8. **Try It 2:** A. Identity Property — *Multiplying by 1 keeps the same value, so this is the Identity Property of Multiplication.*
9. **Exit Ticket:** A. Associative Property — *The grouping (parentheses) changed but the numbers stayed in the same order, so this is the Associative Property of Addition.*